

# Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator

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|-----------|-----------------------------------------|
| Week      | Week 5                                  |
| Track     | BA (Business Analytics)                 |
| Task Type | Simulation & Quantitative Risk Modeling |

## Description

Students must design a multi-asset financial risk simulator that models portfolio behavior under uncertainty and quantifies downside risk using industry-standard quantitative finance techniques.

## Core Requirements

- Compute the **historical covariance matrix** for a universe of 500 assets.
- Execute a **Cholesky decomposition** of the covariance matrix to correlate multi-dimensional random noise.
- Simulate **N independent Geometric Brownian Motion (GBM) paths** for the asset universe.
- Calculate the portfolio's **99% Value-at-Risk (VaR)** from the simulated outcome distribution.

## Suggested Workflow

1. **Data collection** — gather historical price series for the 500-asset universe and derive daily/periodic returns.
2. **Covariance estimation** — build the 500×500 historical covariance matrix from the return series.
3. **Correlation structure** — apply Cholesky decomposition to the covariance matrix to obtain a lower-triangular factor for correlating independent standard normal draws.
4. **Path simulation** — generate N correlated GBM paths per asset using the Cholesky-transformed noise, drift, and volatility estimates.
5. **Portfolio aggregation** — combine simulated asset paths into portfolio-level returns using specified weights.
6. **Risk metric** — compute the 99% VaR (and optionally CVaR/Expected Shortfall) from the simulated portfolio loss distribution.

## Key Concepts

|                                  |                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Covariance Matrix</b>         | Captures pairwise co-movement across all 500 assets; the backbone of the correlation structure.       |
| <b>Cholesky Decomposition</b>    | Factorizes the covariance matrix so independent random draws can be transformed into correlated ones. |
| <b>Geometric Brownian Motion</b> | Standard stochastic process for simulating asset price evolution under drift and volatility.          |
| <b>99% VaR</b>                   | The loss threshold not expected to be exceeded with 99% confidence over the simulation horizon.       |